



10830 - Disclosing the nature of N-emitters in the early Universe: Rest-frame near-IR imaging of RXCJ2248-ID at $z=6.1$

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI - RXCJ2248-ID				
	1	F560W	MIRI Imaging	(1) RXCJ2248-ID-center
	2	F770W	MIRI Imaging	(1) RXCJ2248-ID-center
	3	F1000W	MIRI Imaging	(1) RXCJ2248-ID-center
	4	F1280W	MIRI Imaging	(1) RXCJ2248-ID-center

ABSTRACT

JWST has detected a class of high- z galaxies, termed N-emitters, characterized by their elevated N/O abundances relative to O/H. These galaxies have SFR surface densities close to that of maximal starbursts, and stellar mass surface densities close to those of the densest stellar clusters. The nature of these galaxies based on existing rest-UV/optical NIRCам and NIRSspec data is still puzzling and subject to strong debate. Scenarios involving very massive stars (VMS) in extremely short-lived young and compact ($R_e < 100$ pc) starbursts, or accreting black holes have been invoked to explain the elevated N/O. RXCJ2248-ID belongs to this class of galaxies and represents the best candidate to investigate the nature of N-emitters due to its apparent (magnified) brightness and redshift ($z = 6.1$). This program proposes deep MIRI multiwavelength (F560W to F1280W) simultaneous imaging of the lensed images (magnification $\times 5$ -8) of RXCJ2248-ID, securing for the first time the detection of the rest-frame near-infrared continuum emission in a N-emitter. The extension of the existing NIRCам and NIRSspec data into the near-infrared will provide the key additional SED coverage to reveal the presence of hot dust heated by an AGN, and to identify stellar populations other than the young massive stars, i.e. mature stellar populations from previous bursts. The true nature of RXCJ2248-ID, and that of N-emitters, can not be completely disclosed without covering the rest-frame near-infrared range.

OBSERVING DESCRIPTION

We propose to conduct deep MIRI imaging of RXCJ2248-ID using the F560W, F770W, F1000W and F1280W filters. The primary aim of these observations is to detect, for the first time, the rest-frame near-IR continuum of a N-emitter at $z > 6$. The near-IR emission will be key to determine the star formation history and the nature of the dominant ionizing source in this galaxy.

We request the images to be taken in the FASTR1 readout mode with 100 groups per integration, 2 integrations for F560W and F770W and 7 integrations for F1000W and F1280W filters. This strategy yields to 4500 and 15500 seconds for the F560W and F770W, and F1000W and F1280W filters, respectively. These observations are conducted using 8 dithers with the CYCLING LARGE pattern to optimize the sampling of the MIRI PSF and the correction for residual cosmic rays. We request group observations to occur within 3 days to minimize the impact of possible changes in the

background emission and V3 orientation. The RXCJ2248-ID-center is defined to cover the three lensed images (ID1-ID3) of RXCJ2248-ID within a single pointing.

Proposal 10830 - Targets - Disclosing the nature of N-emitters in the early Universe: Rest-frame near-IR imaging of RXCJ2248-ID at z...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	RXCJ2248-ID-center	RA: 22 48 44.8188 (342.1867450d) Dec: -44 32 2.59 (-44.53405d) Equinox: J2000		
Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies] Extended=NO					

Proposal 10830 - Observation 1 - Disclosing the nature of N-emitters in the early Universe: Rest-frame near-IR imaging of RXCJ2248-...

Observation	Proposal 10830, Observation 1: F560W										Fri Jun 05 20:00:10 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	RXCJ2248-ID-center	RA: 22 48 44.8188 (342.1867450d)								
			Dec: -44 32 2.59 (-44.53405d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	8						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	100	2	1	Dither 1	8	16	4462.264	
Special Requirements	Group Observations 1, 2, 3, 4 within 50 Days										

Proposal 10830 - Observation 2 - Disclosing the nature of N-emitters in the early Universe: Rest-frame near-IR imaging of RXCJ2248-...

Observation	Proposal 10830, Observation 2: F770W										Fri Jun 05 20:00:10 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	RXCJ2248-ID-center	RA: 22 48 44.8188 (342.1867450d)								
			Dec: -44 32 2.59 (-44.53405d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	8						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	100	2	1	Dither 1	8	16	4462.264	
Special Requirements	Group Observations 1, 2, 3, 4 within 50 Days										

Proposal 10830 - Observation 3 - Disclosing the nature of N-emitters in the early Universe: Rest-frame near-IR imaging of RXCJ2248-...

Observation	Proposal 10830, Observation 3: F1000W										Fri Jun 05 20:00:10 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	RXCJ2248-ID-center	RA: 22 48 44.8188 (342.1867450d)								
			Dec: -44 32 2.59 (-44.53405d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies] Extended=NO										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	8						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1000W	FASTR1	100	7	1	Dither 1	8	56	15673.426	
Special Requirements	Group Observations 1, 2, 3, 4 within 50 Days										

Proposal 10830 - Observation 4 - Disclosing the nature of N-emitters in the early Universe: Rest-frame near-IR imaging of RXCJ2248-...

Observation	Proposal 10830, Observation 4: F1280W Fri Jun 05 20:00:10 GMT 2026									
	Diagnostic Status: Warning Observing Template: MIRI Imaging									
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	RXCJ2248-ID-center	RA: 22 48 44.8188 (342.1867450d) Dec: -44 32 2.59 (-44.53405d) Equinox: J2000							
Template	<i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies]</i> <i>Extended=NO</i>									
	Subarray FULL									
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
	1	CYCLING	1	8						LARGE
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time
	1	F1280W	FASTR1	100	7	1	Dither 1	8	56	15673.426
Special Requirements	Group Observations 1, 2, 3, 4 within 50 Days									